

# Umair Mehfooz

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## SUMMARY

Artificial Intelligence undergraduate and Machine Learning Intern at FlyRank AI who builds and ships production software end to end: two live e-commerce stores and an offline business-management app for clients, plus LLM applications with LangChain and FastAPI. Works across Python, TypeScript, Next.js, Supabase/PostgreSQL and data tooling such as Pandas and DuckDB.

## EDUCATION

**Capital University of Science and Technology (CUST)** | Islamabad, Pakistan **2024 – 2028 (Expected)**  
*Bachelor of Science in Artificial Intelligence (BSAI) – Dean's Honor Roll (4 times)*

**Federal Government Sir Syed College** | Rawalpindi, Pakistan **2022 – 2023**  
*Intermediate (FSc Pre-Engineering)*

## EXPERIENCE

**FlyRank AI – Machine Learning Intern** **Jun 2026 – Present**

- Analyzing the signals behind Google search ranking and content discoverability using FlyRank's anonymized search data.
- Querying a ~79M-row search dataset with DuckDB directly from Google Colab, without local downloads.
- Running an end-to-end ML workflow: task framing, data contracts, leakage checks, signal audits, baselines and validation.

**Freelance Full-Stack Developer – Independent Client Work** **Aug 2026 – Present**

- Naavya** (naavya.vercel.app): Launched a production fashion e-commerce store on Next.js, TypeScript and Supabase (PostgreSQL, Auth, RLS) with COD and Safepay checkout, customer accounts, order emails and an admin dashboard.
- Ideal Jewellers** (idealjewellers.vercel.app): Built a luxury jewellery storefront with category filters, bridal collections, search, wishlist and checkout; ported the static prototype to Next.js with the catalogue and admin area on Supabase.
- Aqua Aman**: Built an offline Windows desktop app (Electron, React, SQLite) that runs a water-delivery business: deliveries, payments, bottle stock, P&L and receivables reports, scheduled WhatsApp receipts, automatic daily backups and a PIN lock.

## PROJECTS

**KeepMe – AI Accountability Coach** | Next.js, FastAPI, Supabase, pgvector, OpenRouter, Docker **2026**

- Built a full-stack AI coaching PWA that tracks commitments, extracts structured facts from daily reflections, scores progress, and sends reports and Web Push reminders.
- Computed coaching tone deterministically so the LLM picks the wording, never the severity; replies pass a safety filter.
- Enforced PostgreSQL Row Level Security through FastAPI by verifying Supabase JWTs per request; added PII redaction, AI usage quotas, a circuit breaker and model fallback; deployed with Docker and Caddy.

**BidForge AI – RAG Proposal Engine (Hackathon)** | LangChain, FAISS, FastAPI, Groq API **Jun 2026**

- Built a RAG pipeline matching RFP requirements to capabilities using MiniLM embeddings, FAISS and re-ranking.
- Generated compliance verdicts and proposal drafts with Groq LLMs, cutting drafting time from hours to minutes.

**Portal Automation Framework** | Python, Playwright, Pytest **Jun 2026**

- Built a browser automation framework with the Page Object Model, session reuse, parallel tests and failure screenshots.

## TECHNICAL SKILLS

**Programming Languages:** Python, TypeScript, JavaScript, SQL, C++  
**AI/ML & Data Science:** NumPy, Pandas, Matplotlib, Scikit-learn, DuckDB, Jupyter, Machine Learning, RAG  
**AI & LLM Frameworks:** LangChain, Prompt Engineering, LLM Application Development, OpenRouter, Groq API, pgvector  
**Web & Backend:** Next.js, React, Tailwind CSS, FastAPI, REST APIs, Electron  
**Databases:** PostgreSQL, Supabase, MongoDB, SQLite  
**Tools & Cloud:** Git, GitHub, Docker, Vercel, Playwright, Oracle Cloud Infrastructure (OCI), MS Excel

## CERTIFICATIONS

- Oracle Cloud Infrastructure (OCI) Certified Foundations Associate – Oracle **Aug 2025**
- Python for Data Analysis: Pandas & NumPy – Coursera

## LANGUAGES & INTERESTS

**Languages:** English, Urdu, German (A1 certificate, NUML)  
**Interests:** Book Reading, Gardening, Socializing